From Zeta Zeros to Engineering: What the Cathedral's Mathematical Physics Means for Practicing Engineers

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April 19, 2026

Abstract

The Cathedral project establishes a machine-verified equivalence between the Riemann Hypothesis and the decay of the Nyman–Beurling distance, using techniques from spectral theory, harmonic analysis, and variational principles. While the mathematical content is pure number theory, the *techniques* are drawn from the same toolkit that practicing engineers use daily: Fourier analysis, eigenvalue problems, variational methods, and structured matrix theory. This paper maps the Cathedral's mathematical physics to five engineering domains: signal processing, structural analysis, wireless communications, control theory, and numerical methods. We do not claim that the Cathedral's results have direct engineering applications—they do not. Instead, we argue that the *mathematical structures* that appear in the Cathedral illuminate engineering techniques from an unexpected angle, and that engineers may find pedagogical and conceptual value in seeing their daily tools applied to a 167-year-old unsolved problem.

Contents

1 Introduction: The Engineer's Toolkit in Disguise

Engineers solve problems with a canonical set of mathematical tools:

- Fourier transforms and spectral analysis.
- Eigenvalue problems and matrix decompositions.
- Variational principles and optimization.
- Structured matrices (Toeplitz, Hankel, Gram).
- Digital filtering and windowing functions.

The Cathedral uses every one of these tools, applied to a problem that has nothing to do with engineering—the distribution of prime numbers. This paper makes the connection explicit, showing engineers exactly where their tools appear in the proof architecture and what the number-theoretic results look like through an engineering lens.

1.1 What This Paper Is (and Isn't)

This paper is: A translation guide that maps Cathedral mathematics to engineering concepts, for engineers who are curious about what a "proof about primes" looks like when you strip away the number theory.

This paper is not: A claim that the Cathedral helps engineers build better bridges, design better antennas, or write better DSP code. The connection is structural (same tools) not functional (same applications).

2 The Parseval Bridge as Signal Processing

2.1 The Engineering Parseval Theorem

Every electrical engineer learns Parseval's theorem in their first signals course:

$$\int_{-\infty}^{\infty} |x(t)|^2 \, dt = \int_{-\infty}^{\infty} |X(f)|^2 \, df.$$

The energy of a signal in the time domain equals the energy in the frequency domain. This is the foundation of spectral analysis, power spectral density, and filter design.

2.2 The Cathedral's Version

The Cathedral proves a specific instance:

$$\int_0^1 |1 - f_N(x)|^2 \, dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\hat{M}_N(1/2 + it)|^2 \, dt,$$

where $f_N(x)$ is a linear combination of fractional-part functions $\{1/(kx)\}$, and $\hat{M}_N(s)$ is its Mellin transform on the critical line.

Engineering translation:

Cathedral	Engineering
$\int_0^1 \|1 - f_N\|^2 \, dx$	Time-domain error energy
$\hat{M}_N(1/2 + it)$	Frequency-domain spectrum
Mellin transform	Laplace-like transform ($x = e^{-u}$)
Critical line $\mathrm{Re}(s) = 1/2$	Imaginary axis ($j\omega$)
$d_N^2 \to 0$	Error energy $\to 0$
RH	The signal can be perfectly reconstructed

The Mellin transform is related to the bilateral Laplace transform by the substitution $x = e^{-u}$:

$$\mathcal{M}[f](s) = \int_0^{\infty} f(x)\, x^{s-1} \, dx = \int_{-\infty}^{\infty} f(e^{-u})\, e^{-su} \, du = \mathcal{L}[f(e^{-(\cdot)})](s).$$

So the "critical line" $\mathrm{Re}(s) = 1/2$ in number theory corresponds to the imaginary axis $s = 1/2 + j\omega$ in the s-plane—exactly where control engineers evaluate transfer functions for stability (the Nyquist criterion).

2.3 The Window Function Connection

The Cathedral's optimal test vectors v_k are the Selberg sieve weights:

$$v_k = \mu(k) \left(1 - \frac{\ln k}{\ln N}\right)^+,$$

where $\mu(k)$ is the Möbius function. The factor $(1 - \ln k / \ln N)^+$ is a triangular window in log-space.

Engineering translation: This is a *Bartlett window* applied to the Möbius spectrum. Every DSP engineer knows Bartlett—it's the simplest window that tapers to zero at the edges, reducing spectral leakage at the cost of widening the main lobe.

The Cathedral's numerical experiments show this Bartlett window achieves $Q_N \sim 12.45 \ln N$, while the optimal window achieves $Q_N \sim 21.65 \ln N$. The ratio $12.45/21.65 \approx 0.575$ is analogous to the efficiency loss from using a Bartlett window instead of a Dolph–Chebyshev or Kaiser window in DSP.

Open question for engineers: What is the optimal window function $w(k)$ that minimizes d_N^2 for a given N? This is equivalent to the optimal sidelobe suppression problem in antenna array design.

3 The Gram Matrix as a Structural Problem

3.1 Positive Definiteness and Stability

The Cathedral proves that the augmented Gram matrix H_N is positive definite for all $N \geq 1$. In engineering:

- **Structural engineering**: A positive definite stiffness matrix K means the structure has no zero-energy deformation modes—it is stable.
- **Control theory**: A positive definite Lyapunov matrix P certifies stability of a linear system $\dot{x} = Ax$.
- **Machine learning**: A positive definite kernel matrix K means the kernel is valid (Mercer's theorem).

The Cathedral's proof that H_N is PD uses the "Factorial Nuke": the augmented entries involve $1/k!$, which decay so rapidly that they dominate the spectral structure regardless of the off-diagonal entries. This is standard in numerical analysis: diagonal dominance implies positive definiteness.

3.2 The Rayleigh Quotient

The Rayleigh quotient $R(v) = v^T Gv/v^T v$ gives the minimum eigenvalue:

$$\lambda_{\min}(G) = \min_{v \neq 0} R(v).$$

Engineers use this in:

- **Vibration analysis**: The smallest eigenvalue of the generalized eigenvalue problem $Kv = \lambda Mv$ gives the fundamental natural frequency.
- **Buckling analysis**: The smallest eigenvalue of $K_g v = \lambda Kv$ gives the critical buckling load.
- **Principal Component Analysis**: The eigenvalues of the covariance matrix determine variance explained.

The Cathedral's $\lambda_{\min}(G_N) \sim C/\ln N$ (where C is the quantum stiffness constant) means the Gram matrix becomes increasingly ill-conditioned as N grows—but never singular. Engineers encounter exactly this behavior in mesh refinement for finite element analysis: finer meshes produce larger, more ill-conditioned stiffness matrices, but the physics ensures they remain positive definite.

4 Spectral Gaps and Network Engineering

4.1 Algebraic Connectivity

In graph theory, the second-smallest eigenvalue of the Laplacian matrix L is the *algebraic connectivity* (Fiedler value). It measures how well-connected a network is:

- $\lambda_2 = 0$: disconnected graph.
- $\lambda_2 > 0$: connected, with larger values meaning better connectivity.
- λ_2 controls mixing time of random walks on the graph.

The Cathedral's "spectral gap" $\lambda_{\min}(G_N) > 0$ for all N plays an analogous role: it ensures that the Gram matrix is always invertible, meaning the best L^2 approximation always exists and is unique.

4.2 MIMO Channel Matrices

In wireless communications, the channel capacity of a MIMO system depends on the eigenvalues of HH^*, where H is the channel matrix. The eigenvalue distribution (Marchenko–Pastur for random channels) determines how many independent data streams can be multiplexed.

The Gram matrix G_N is *not* random—its entries are deterministic functions of $\gcd(j,k)$. But its eigenvalue distribution shares qualitative features with random matrix theory: the bulk eigenvalues grow linearly while the minimum eigenvalue grows logarithmically. This "soft edge" behavior is reminiscent of the Tracy–Widom distribution at the edge of random matrix spectra.

5 Variational Methods: From RH to Finite Elements

5.1 The Ritz Method

The Cathedral finds the optimal test vector by minimizing:

$$d_N^2 = \min_{v\in\mathbb{R}^{N-1}} \int_0^1 \left(1 - \sum_{k=2}^{N} v_k\{1/(kx)\}\right)^2 dx = 1 - b^T G^{-1} b.$$

This is a *Ritz approximation*: minimize the energy functional over a finite-dimensional subspace.

Engineers use exactly this structure in:

- **Finite Element Method**: Minimize $\Pi(u) = \frac{1}{2}u^T K u - u^T f$ over the span of shape functions.
- **Method of Moments**: Project Maxwell's equations onto a finite basis to compute antenna patterns.
- **Galerkin methods**: Approximate PDE solutions by minimizing residuals in a Hilbert space.

The Cathedral's "basis functions" are $\{1/(kx)\}$ for $k = 2, \ldots, N$. The "target function" is $\mathbf{1}_{(0,1)}$ (the constant function 1). The "stiffness matrix" is G_N. The "load vector" is $b_k = \int_0^1 \{1/(kx)\}\, dx$.

The entire Nyman–Beurling criterion reduces to: *as the mesh refines ($N \to \infty$), does the FEM solution converge to the exact solution?* RH says yes.

5.2 Convergence Rate and Mesh Quality

In FEM, the convergence rate depends on:

1. The regularity of the target function (smoothness).
2. The quality of the mesh (element aspect ratios).
3. The condition number of the stiffness matrix.

For the Cathedral:

1. The target is **1**—infinitely smooth.
2. The "mesh" is the set of integers $\{2, \ldots, N\}$—with arithmetic structure instead of geometric structure.
3. The condition number $\kappa(G_N) \sim N \ln N$—growing, but controlled.

The convergence rate $d_N^2 \sim C/\ln N$ (under RH) is *logarithmic*—much slower than polynomial FEM convergence rates. This reflects the difficulty of the approximation problem: the basis functions $\{1/(kx)\}$ are arithmetically structured, not designed for optimal approximation.

6 Control Theory: Stability on the Critical Line

6.1 The Nyquist Connection

In classical control theory, a system with transfer function $G(s)$ is stable if no poles lie in the right half-plane $\mathrm{Re}(s) > 0$. The Nyquist stability criterion evaluates $G(j\omega)$ on the imaginary axis to determine stability.

The Riemann Hypothesis asserts that the nontrivial zeros of $\zeta(s)$ lie on the line $\mathrm{Re}(s) = 1/2$—the "critical line." Under the substitution $s \mapsto s + 1/2$, this becomes: all zeros lie on the imaginary axis.

Translation: RH says that a certain "transfer function" (related to $1/\zeta(s)$, the Möbius generating function) has all its zeros on the imaginary axis—the boundary of stability. The primes are a marginally stable system.

6.2 The Spectral Zeta Function as a Transfer Function

Define $F(s) = \sum_{n=1}^{\infty} \mu(n)n^{-s}$. Then $F(s) = 1/\zeta(s)$, and:

- RH $\iff$ all zeros of $F(s)$ have $\mathrm{Re}(s) = 1/2$.
- $F(s)$ converges absolutely for $\mathrm{Re}(s) > 1$ (the "high-frequency" region).
- The behavior on $1/2 < \mathrm{Re}(s) < 1$ is the "critical strip"—the frequency band where the system's behavior is uncertain.

This is reminiscent of robust control: we know the system is stable at high frequencies, but the critical behavior is in a finite bandwidth around the stability boundary.

7 Numerical Methods: Condition Numbers and the Quantum Stiffness

7.1 The Condition Number Problem

Numerical engineers care about the condition number $\kappa(A) = \|A\| \cdot \|A^{-1}\|$, which controls the amplification of roundup errors in solving $Ax = b$.

The Cathedral's Gram matrix has condition number $\kappa(G_N) \sim N \ln N/C$, where $C \approx 0.046$ is the quantum stiffness constant. This is a moderately ill-conditioned system: worse than a Hilbert matrix ($\kappa \sim e^{3.5N}$) but much better than the Vandermonde matrix ($\kappa \sim N^N$).

Engineering implication: The Gram matrix can be solved numerically in double precision for $N \lesssim 10^{12}$ before roundoff destroys the solution. This is why the Cathedral's Rust experiments use exact rational arithmetic—to avoid conditioning issues entirely.

7.2 Iterative Methods

The Gram matrix is symmetric positive definite, which makes it ideal for the Conjugate Gradient method. The convergence rate depends on $\sqrt{\kappa}$:

$$\|x_k - x_*\| \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k \|x_0 - x_*\|.$$

For $\kappa \sim N \ln N$, this gives convergence in $O(\sqrt{N \ln N})$ iterations—manageable for moderate N.

8 The Dictionary

Cathedral	**Engineering Domain**	**Engineering Concept**
Parseval Bridge	Signal processing	Time-frequency duality
$d_N^2 \to 0$	DSP	Error energy convergence
Selberg weights	DSP	Bartlett window function
Gram matrix G_N	Structural / FEM	Stiffness matrix
$\lambda_{\min}(G_N)$	Structural / FEM	Fundamental frequency
H_N positive definite	Control	Lyapunov stability
Rayleigh quotient	Vibrations	Natural frequency bound
Critical line $\mathrm{Re}(s) = 1/2$	Control	Nyquist imaginary axis
$1/\zeta(s)$	Control	Transfer function
RH	Control	Marginal stability
$\kappa(G_N)$	Numerical methods	Condition number
$Q_N / \ln N \to C$	Materials	Spectral stiffness constant
Ritz approximation	FEM	Galerkin projection
Axiom elimination	V&V	Assumption retirement

9 What Engineers Cannot Learn from the Cathedral

For intellectual honesty:

1. The Cathedral does not provide new signal processing algorithms. The Parseval Bridge is a specific application of a known theorem.

2. The Gram matrix structure (entries involving gcd and lcm) does not appear in typical engineering problems. No bridge or antenna produces a Gram matrix.

3. The convergence rate $d_N^2 \sim C/\ln N$ is far too slow for any engineering approximation scheme. Engineers would never accept logarithmic convergence.

4. The "quantum stiffness constant" $C \approx 0.046$ is a number-theoretic constant with no known physical realization.

The value for engineers is *conceptual*, not *practical*: seeing familiar tools applied to an unfamiliar problem can deepen understanding of why those tools work and what their fundamental limitations are.

10 Conclusion

The Cathedral is pure mathematics, verified by machine. But its tools are the engineer's tools: Fourier analysis, eigenvalue problems, variational methods, condition numbers, and window functions. The primes are not a bridge or a circuit or an antenna, but they respond to the same mathematical forces.

For the practicing engineer, the Cathedral offers a mirror: *the mathematics you use to design filters and analyze structures is the same mathematics that governs the distribution of prime numbers.* This is not a coincidence. It is a reflection of the universality of harmonic analysis—the deepest tools work everywhere, from the critical line of the zeta function to the critical frequency of a low-pass filter.

The engineer's toolkit was built for bridges.
It turns out the primes are also a bridge—
between order and chaos, between structure and randomness.
The tools work because the bridge is real.

Source code: `https://github.com/jrgochan/prime`

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